



Delivery

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Difficulty Level : easy

Time Limit: 0.3 s

Description

In Oued Smar city, deliveries are handled with speed and precision. Among the team of delivery workers is Karim, a dedicated delivery man known for his efficiency. Today, Karim is on a mission to deliver packages to customers as quickly as possible while choosing the best delivery method.

Karim has two delivery options:

1. Manual Delivery: Personally queue up at the central dispatch hub. He must wait for all

ongoing deliveries to be processed before it's his turn. Once done, he takes " x " time to deliver the package.

2. Automated Delivery: Submit the package to an automated system that only begins dispatching once a specific number " m " of deliveries have been completed. After that, the package takes a fixed time " y " to reach the destination.

The system currently has a list of deliveries in queue, each taking a known amount of processing time. For each delivery scenario Karim faces, you're given the options and parameters. Your job is to help Karim decide which method will get the package to the customer in the shortest time.



Input

- The first line contains a single integer n ($1 \leq n \leq 10^5$): the number of deliveries currently in the queue.
- The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$): the time to process each delivery.
- The third line contains an integer q ($1 \leq q \leq 10^5$): the number of queries.
- The next q lines each contain three integers x, y , and m ($1 \leq x, y \leq 10^9$ and $1 \leq m \leq n$).
 - x : Time taken for manual delivery.
 - y : Time taken for the automated delivery.
 - m : The number of deliveries that must be completed before the automated system sends the package.

Output

- For each query, output a single line containing a single integer: the minimum time required to deliver the package.

Examples

Input	Output
5	8
1 2 1 3 4	13
5	7
3 4 3	7
5 10 2	21
1 3 3	
10 4 2	
10 15 5	



Note: It is guaranteed that the number of deliveries to process in each query m does not exceed n .